

Population Distributions of Footbridge Modal Properties: A PRISMA-Guided Review and Hierarchical Bayesian Analysis

Samadhi Loku Hetti Arachchilage^a, Colin Caprani^a, Stana Živanović^b

^a*Department of Civil Engineering, Monash University, Melbourne, Victoria, Australia*

^b*School of Engineering, University of Warwick, Coventry, United Kingdom*

Abstract

Keywords:

1. Introduction

The increasing adoption of light weight materials and slender structural forms has made modern footbridges more susceptible to human-induced vibrations. As a result, vibration serviceability under pedestrian loading has become an important design consideration to ensure acceptable structural performance and user comfort. Such assessment depends on several aspects of the bridge, including its structural form, material properties, and dynamic characteristics [1, 2, 3]. Therefore, prior knowledge or reliable estimation of key modal parameters is essential at the design stage, as these parameters provide an initial basis for assessing the feasibility of a proposed design [4, 5, 6, 7].

The current vibration serviceability codes and guidelines represent pedestrian loading through simplified load models to enable practical application at the design stage [8, 9, 10, 11, 12, 13, 14]. Since these load models are used to assess the vibration response of a wide range of footbridges, their adequacy should be evaluated in terms of the resulting structural response, rather than solely through how accurately they reproduce pedestrian loading.

The key limitation of many existing simplified load models is that pedestrians are represented as moving forces, with the corresponding load models calibrated without explicitly accounting for the dynamic interaction between the pedestrians and the supporting structure [15, 16, 17]. Human-structure interaction (HSI) alters the dynamic properties of the coupled pedestrian-structure system, and can lead to structural responses that differ substantially from those predicted by simplified load models. These effects can be

31 particularly important for lightweight structures and for cases with relatively
32 large pedestrian-to-structure mass ratios [18, 17]. Consequently, simplified
33 code models calibrated without accounting for HSI may exhibit systematic
34 response bias that varies with the dynamic characteristics of the bridge, mo-
35 tivating their reassessment and response-based recalibration.

36 Response-based calibration provides such a framework by comparing the
37 response predicted by the simplified design model with that obtained from ex-
38 perimental observations or a higher-fidelity reference model over a prescribed
39 range of structural properties. Previous studies have adopted this approach
40 by evaluating simplified pedestrian loading or vibration serviceability mod-
41 els over ranges of natural frequency, damping and other structural param-
42 eters [19, 3, 17, 18, 20]. However, these structural domains considered in
43 such studies have generally been defined using prescribed parameter ranges,
44 idealised structural configurations, or regularly sampled parameter combi-
45 nations rather than using the observed statistical distribution of dynamic
46 properties within the footbridge population. As a result, some structural
47 conditions may receive disproportionate importance in the calibration sim-
48 ply because of the adopted sampling scheme, rather than because they are
49 common in practice.

50 For code-model calibration, it is important to know not only the range
51 of structural properties that may occur, but also how common these proper-
52 ties are within the footbridge population. Several recent studies have made
53 valuable efforts to compile databases of footbridge characteristics and modal
54 properties. Oliveira et al. (2014) [21] reported fundamental frequencies
55 for 79 footbridges in Portugal, while Van Nimmen (2015) reviewed data
56 for more than 160 footbridges [22, 23]. Wei et al. (2019) [24] assembled
57 dynamic-property data for 138 footbridges, and Gong et al. (2024) [25]
58 more recently studied a collection of footbridges in China. These studies
59 provide a substantial starting point, but their datasets were assembled for
60 different purposes, report different properties, and include some of the same
61 bridges. Further observations are dispersed across many individual case stud-
62 ies [26, 6, 27, 20]. Consolidating these records, identifying duplicate bridges,
63 and recording which modal properties are available would provide a clearer
64 basis for estimating their observed distributions.

65 This study addresses the above gap in two stages. First, a systematic
66 literature search and screening procedure following the PRISMA framework
67 is used to identify relevant publications and compile bridge-level data on key
68 modal properties. Second, hierarchical Bayesian regression is used to proba-
69 bilistically characterise incompletely observed modal properties while retain-
70 ing the uncertainty associated with their estimation. The resulting database
71 provides a population-level description of key footbridge modal properties

and their relationships, allowing the prevalence of different combinations of natural frequency, damping ratio and modal mass to be quantified. Such information provides a statistical basis for distinguishing commonly occurring structural conditions from those that are comparatively rare. This, in turn, can support the selection and weighting of representative structural parameter combinations used in vibration-serviceability assessment and code-model calibration.

Accordingly, this study aims to characterise the modal properties of the footbridge population using a systematically assembled bridge-level database of approximately 250 structures. The principal contributions of the study are: (i) the systematic identification and consolidation of published footbridge dynamic-property data; (ii) the construction of a large, non-duplicated database of bridge-level modal properties; (iii) the probabilistic treatment of incompletely observed damping and modal-mass information using hierarchical Bayesian regression; and (iv) the population-level characterisation of modal properties relevant to vibration-serviceability assessment and code-model calibration.

2. Systematic database development

The footbridge database was developed through a systematic literature identification and screening process following the PRISMA framework [28]. The procedure includes publication identification, title and abstract screening, full text eligibility assessment, bridge level data extraction and consolidation of information into a single database. The following subsections describe the search and screening procedure, the data extraction process, and the resulting database.

2.1. Review scope and search strategy

The systematic review aimed to identify published investigations reporting structural and dynamic properties of full scale pedestrian bridges. The principal variables of interest were bridge span, deck width, fundamental vertical natural frequency, vertical damping ratio, Modal mass, structural form, cross section, and construction materials.

Scopus was selected as the literature database because of its broad coverage of peer-reviewed engineering research [29]. The candidate publications were identified using Elsevier Scopus search API through title, abstract, and keyword fields. The search used a broader combination of pedestrian-bridge terminology with terms relating to natural frequency, damping, and structural mass, experimental dynamic testing, ambient vibration testing, operational modal analysis, model updating and vibration serviceability.

110 The search returned 727 bibliographic records before deduplication. Dub-
111 plicate records were identified using DOI, Scopus EID. Following deduplica-
112 tion, 697 unique records were retained for screening. A DOI was available for
113 529 records, while Scopus EID were retained for publications without a DOI.
114 Scopus records were used to identify and screen publications, while numerical
115 bridge properties were extracted from the full texts of the eligible studies.

116 *2.2. Eligibility criteria*

117 Publications were considered eligible when they reported an investigation
118 of an existing, full-scale pedestrian bridge and contained experimentally mea-
119 sured or identified structural or modal properties that could be associated
120 with a specific structure. Studies based only on laboratory specimens, scaled
121 models, hypothetical structures, or purely numerical investigations were ex-
122 cluded. Publications were also excluded where the reported frequency could
123 not be identified as a structural vertical natural frequency relevant to the
124 present study. For inclusion in the bridge-level database, sufficient informa-
125 tion was required to identify the bridge and extract its fundamental vertical
126 natural frequency together with the geometric information required for the
127 population analysis. Damping ratio, modal mass, structural form, cross-
128 section, and construction material were extracted where available but were
129 not mandatory for inclusion.

130 *2.3. Study screening and bridge identification*

131 Following bibliographic deduplication, the 697 unique publications were
132 retained for full-text screening. The full-text documents were processed
133 using a semi-automated Python workflow. PDF text was extracted using
134 PyMuPDF and subsequently searched for evidence of full-scale pedestrian
135 bridge investigations and relevant structural and modal properties.

136 The extracted text was searched for bridge names and terminology as-
137 sociated with natural frequency, damping, modal mass, mode shapes, and
138 experimental vibration testing. Reference sections were excluded from the
139 search to reduce the likelihood of identifying bridges mentioned only in cited
140 literature.

141 During the screening stage, each publication was counted as a single
142 record, regardless of how many bridges it investigated. During data ex-
143 traction, however, each individual bridge was treated as a separate database
144 record. Consequently, a single publication could contribute data for several
145 bridges.

146 Each detected bridge was assigned a separate candidate record containing
147 its source publication, bridge identity, and associated structural and modal-
148 property evidence. Candidate records were subsequently reviewed against

149 the bridge-level inclusion requirements defined in Section 2.2 before consoli-
150 dation.

151 *2.4. Bridge-level data extraction and consolidation*

152 For each candidate bridge, the extracted information included bridge
153 name and location, span, deck width, fundamental vertical natural frequency,
154 vertical damping ratio, modal mass, structural form, cross-section, and con-
155 struction material. Numerical quantities were converted to consistent units
156 before consolidation.

157 Because the same physical bridge could be reported in several publica-
158 tions, repeated bridge records were identified using bridge name, location,
159 geometry, structural characteristics, reported modal properties, and source
160 references. Potential matches across different publications were reviewed
161 manually rather than being automatically removed.

162 Where several publications referred to the same physical bridge, the avail-
163 able information was consolidated into a single bridge-level record. The orig-
164 inal experimental investigation was retained as the principal source where
165 available, while additional information from later publications was incorpo-
166 rated where appropriate.

167 *2.5. Resulting database and data completeness*

168 Following bridge-level screening and duplicate reconciliation, 248 unique
169 footbridges were retained in the final database. Fundamental vertical natural
170 frequency and span were available for all bridges, while deck width was avail-
171 able for 240 bridges. Vertical damping ratios were reported for 102 bridges,
172 whereas mass related information was considerably less complete, with modal
173 mass information available for only 30 bridges and linear mass available for
174 25 bridges. Structural form and material information were available for all
175 bridges, while cross-section information was available for 242 bridges. The
176 resulting differences in data completeness, particularly for damping and mass-
177 related properties, motivated the probabilistic treatment introduced in the
178 following section.

179 **3. Bayesian treatment of incomplete bridge data**

180 **4. Discussion**

181 **5. Conclusions**

182 **6. Acknowledgements**

183 This research is supported by an Australian Government Research Train-
184 ing Program (RTP) stipend Scholarship and Monash International Tuition
185 Scholarship (MITS).

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